

Harjas Partap Singh Romana

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EDUCATION

Vellore Institute of Technology

Bachelor of Technology in Computer Science and Engineering | **CGPA: 8.84/10.00**

Bhopal, MP

Sep. 2022 – May 2026

- **Coursework:** Distributed Systems, Data Structures & Algorithms, Database Management Systems, Computer Networks, Object Oriented Programming

EXPERIENCE

QuantAI Limited

Data Scientist Intern

Remote, NZ

May 2025 – Oct. 2025

- Architected an high-concurrency bidding engine using React and Firebase, engineering a Redis-backed publish-subscribe pipeline to process concurrent streams and eliminate data-layer race conditions.
- Optimized a Retrieval-Augmented Generation (RAG) pipeline for an AI Voice Agent by refactoring vector query mechanisms, decreasing inference latency by 5% and improving semantic retrieval accuracy by 2%.
- Engineered a scalable, modular AI Skills Accessor utilizing Flask REST APIs and React, automating assessment generation and reducing manual evaluation workflows by 30%.

PROJECTS

Chronos: Distributed Fault-Tolerant Task Orchestrator | Java, Spring Boot, PostgreSQL, Redis

- Engineered a distributed task orchestration engine to guarantee exactly-once execution semantics across concurrent nodes, solving the double-execution problem for time-sensitive background jobs.
- Architected a two-layer synchronization model, implementing distributed leader election via Redis (Lettuce) to ensure a single active scheduler, and utilizing PostgreSQL `FOR UPDATE SKIP LOCKED` to allow multi-worker queue draining.
- Designed a self-healing fault-tolerance pipeline, featuring a 'Janitor' service to auto-recover zombie tasks from crashed nodes and an exponential backoff algorithm to prevent thundering herd scenarios.

API Rate Limiter & Security Gateway | Java, Redis, Docker

- Architected a distributed Sliding Window API Rate Limiter in Java to protect backend microservices from burst traffic, implementing client backpressure and standardizing HTTP 429 responses.
- Eliminated distributed race conditions by offloading state-management to Redis, executing atomic Lua scripts against Sorted Sets to guarantee absolute thread safety without distributed lock latency overhead.
- Validated robust concurrency controls via highly threaded JUnit stress testing frameworks utilizing `ExecutorService`, sustaining 1,000+ simultaneous requests with zero dropped limits.

CodeAura: Semantic Static Analysis Tool | Node.js, ChromaDB, RAG

- Developed a semantic static analysis engine utilizing ChromaDB as a vector database, enabling deep natural language querying across large-scale codebases to accelerate developer onboarding.
- Engineered a rapid-retrieval indexing pipeline capable of parsing and generating vector embeddings for over 1,000 complex functions, achieving highly relevant query resolution with sub-second latency.

TECHNICAL SKILLS

Languages: Java, Python, C/C++, JavaScript, SQL

Databases & Caching: PostgreSQL, MySQL, Redis, MongoDB, ChromaDB

Cloud & Infrastructure: AWS (EC2, S3, ElastiCache, Auto Scaling, Load Balancing), Docker, CI/CD, GitHub Actions

Architecture & Paradigms: Distributed Systems, System Design, Microservices, REST APIs, RAG

ACHIEVEMENTS & LEADERSHIP

- **Publications:** Co-Authored Springer research paper on Hyperspectral Image Classification, demonstrating advanced analytical and mathematical modeling capabilities.
- **Industry Conclave Finalist (2024):** Analyzed system reliability failures in large-scale distributed architectures and proposed optimization strategies for fault-tolerance.
- **Open Source:** Open Source Contributor for GirlScript Summer of Code (2024), gaining exposure to large-scale collaborative development environments.
- **Leadership:** President of Stats-O-Locked Club & Design Lead for Linux Club, spearheading technical initiatives, community engagement, and student mentorship.